



*Flow control
(if, else and elif)*

➤ Flow Control (if, else, elif)

📖 Flow control structures in Python help in **decision-making** by controlling the **execution flow** of code based on conditions.

Python provides three main conditional statements:

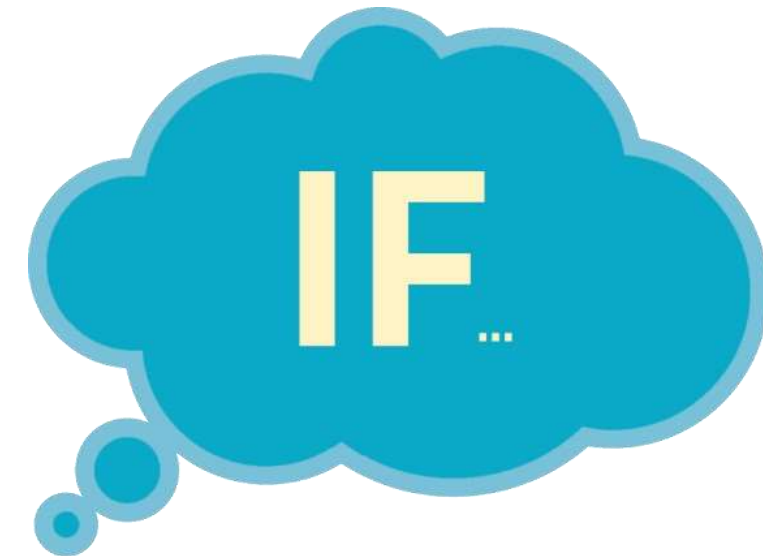
- **if** – executes a block **if the condition is True**
- **elif** – checks **another condition** if the previous one was False
- **else** – executes if **no previous condition** was True

✓ **Syntax:**

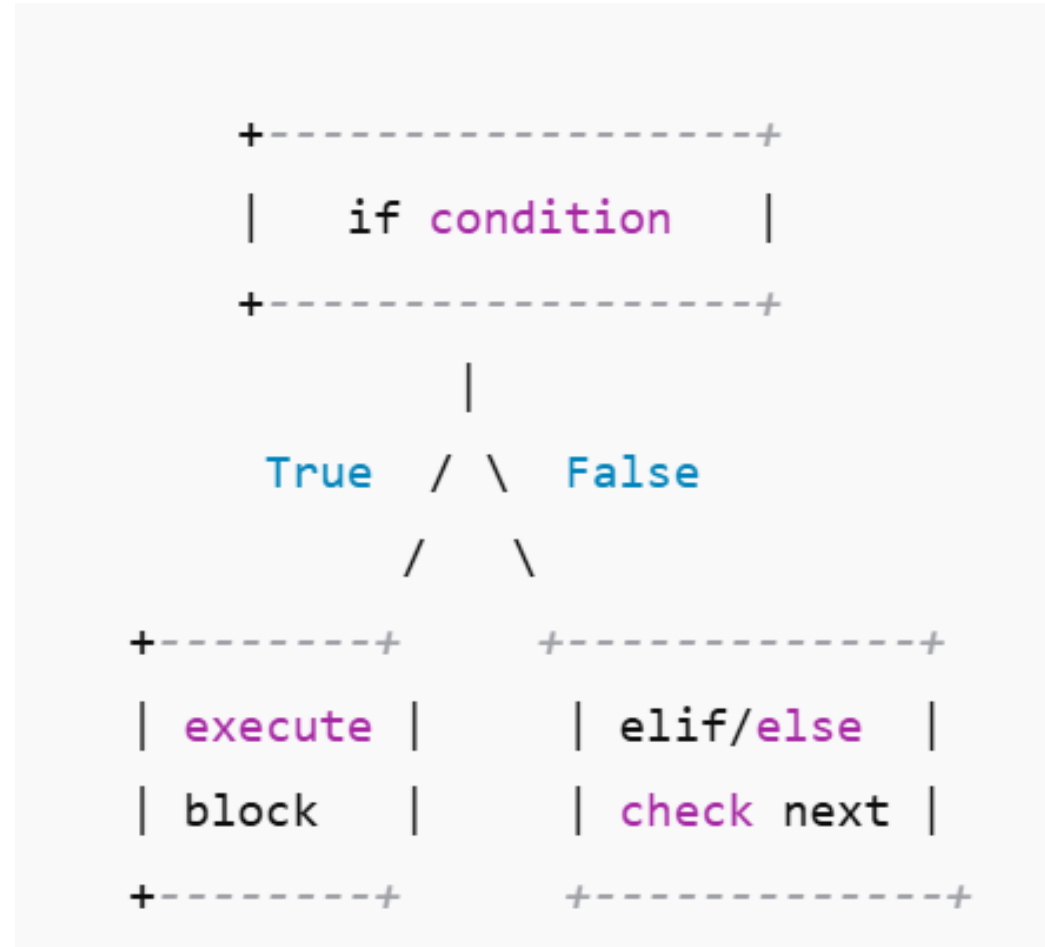
```

hello.py
1  if condition:
2      # block of code if condition is True
3  elif another_condition:
4      # block of code if another_condition is True
5  else:
6      # block of code if none of the above are True
7

```



Flowchart:



➤ 🔍 Example 1: Simple if-else

```
hello.py
1  age = 20
2
3  if age >= 18:
4      print("Eligible to vote")
5  else:
6      print("Not eligible to vote")
7
```

Output:

```
C:\Users\RITESH\Desktop\
Eligible to vote
```



➤ 🔍 Example 2: if-elif-else

```
hello.py > ...  
1  marks = 75  
2  
3  if marks >= 90:  
4      print("Grade A")  
5  elif marks >= 70:  
6      print("Grade B")  
7  elif marks >= 50:  
8      print("Grade C")  
9  else:  
10     print("Grade D")  
11
```

Output:

```
C:\Users\RITESH\Desktop\  
Grade B
```





Key Points:

- Conditions must evaluate to **True or False**
- Python uses **indentation** to define code blocks
- You can have **multiple elif**, but **only one if** and **one else**
- Conditions are checked **top-down**



Real-Life Use Cases:

Authentication System:

```
hello.py > ...  
1  username= input("Enter usernamae: ")  
2  password= input("Enter password: ")  
3  if username == "admin" and password == "1234":  
4      print("Access granted")  
5  else:  
6      print("Access denied")
```

Output:

```
C:\Users\RITESH\Desktop\unn\JAVA>python hello.py  
Enter usernamae: ritesh  
Enter password: 12345  
Access denied
```

```
C:\Users\RITESH\Desktop\unn\JAVA>python hello.py  
Enter usernamae: admin  
Enter password: 1234  
Access granted
```



➤ Real-Life Use Cases:

Traffic Light Logic:

```
hello.py > ...  
1  color = input("Enter color: ")  
2  if color == "red":  
3      print("Stop")  
4  elif color == "yellow":  
5      print("Wait")  
6  elif color == "green":  
7      print("Go")  
8  else:  
9      print("Invalid color")  
10
```

Output:

```
C:\Users\RITESH\Desktop\u  
Enter color: red  
Stop  
  
C:\Users\RITESH\Desktop\u  
Enter color: green  
Go  
  
C:\Users\RITESH\Desktop\u  
Enter color: pink  
Invalid color
```



➤ Real-Life Use Cases: Billing and Discounts:

```
hello.py > ...  
1  total= int(input("Enter total: "))  
2  discount = 0  
3  if total > 1000:  
4      |   discount = 20  
5  elif total > 500:  
6      |   discount = 10  
7  else:  
8      |   discount = 5  
9  print(discount)
```

Output:

```
C:\Users\RITESH\Desktop\  
Enter total:  2000  
20  
  
C:\Users\RITESH\Desktop\  
Enter total:  600  
10
```



Real-Life Use Cases:

Check Even or Odd

```
num = 42

if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

Output:

Even number



Real-Life Use Cases:

Find the Largest of Three Numbers

```
a = 15
b = 27
c = 9

if a > b and a > c:
    print("A is the largest")
elif b > a and b > c:
    print("B is the largest")
else:
    print("C is the largest")
```

Output:

```
B is the largest
```





Real-Life Use Cases:

Nested if Statement

```
score = 85

if score >= 50:
    if score >= 75:
        print("Distinction")
    else:
        print("Pass")
else:
    print("Fail")
```

Output:

Distinction





Real-Life Use Cases:

Multiple Conditions using and and or

```
age = 25
has_license = True

if age >= 18 and has_license:
    print("Can drive")
else:
    print("Cannot drive")
```

Output:

Can drive



Real-Life Use Cases:

Temperature Checker

```
temp = 32

if temp > 35:
    print("Very Hot")
elif temp > 25:
    print("Warm")
elif temp > 15:
    print("Cool")
else:
    print("Cold")
```

Output:

Warm



Real-Life Use Cases:

Simple Calculator (Using Flow Control)

```
a = 10
b = 5
operator = '*'

if operator == '+':
    print("Addition:", a + b)
elif operator == '-':
    print("Subtraction:", a - b)
elif operator == '*':
    print("Multiplication:", a * b)
elif operator == '/':
    print("Division:", a / b)
else:
    print("Invalid operator")\
```

Output:

Multiplication: 50



Real-Life Use Cases:

Grade Calculator

```
marks = 72

if marks >= 90:
    print("Grade: A+")
elif marks >= 75:
    print("Grade: A")
elif marks >= 60:
    print("Grade: B")
elif marks >= 50:
    print("Grade: C")
else:
    print("Fail")
```

Output:

Grade: A





Real-Life Use Cases:

Leap Year Checker

```
year = 2024

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Output:

```
Leap Year
```





Real-Life Use Cases:

Password Strength Checker

```
password = "hello@123"

if len(password) < 6:
    print("Password too short")
elif not any(char.isdigit() for char in password):
    print("Add a number")
elif not any(char in "@# $" for char in password):
    print("Add a special character")
else:
    print("Strong password")
```

Output:

```
Strong password
```



Real-Life Use Cases:

Ticket Price Based on Age

```
age = 70

if age < 5:
    print("Free Entry")
elif age < 18:
    print("Child Ticket: ₹50")
elif age < 60:
    print("Adult Ticket: ₹100")
else:
    print("Senior Ticket: ₹70")
```

Output:

Senior Ticket: ₹70



Real-Life Use Cases:

ATM Withdrawal Validation

```
balance = 5000
withdraw = 2000

if withdraw <= balance:
    balance -= withdraw
    print("Withdrawal successful. New balance:", balance)
else:
    print("Insufficient balance")
```

Output:

```
Withdrawal successful. New balance: 3000
```



